BOUBAKAR DIALLO

Quincy, MA 02171 | 857·308·6457 | bd453@cornell.edu | linkedin.com/in/boubakar | github.com/boubakar12 | boubakar12.github.io

EDUCATION

Cornell University **B.S. Electrical & Computer Engineering** **May 2028**

Honors: Community Leadership Award (Cornell University 2025) **GPA: 3.7**

Coursework: Embedded Systems, Digital Logic & Computer Organization, Circuits, Computer Systems Programming, Linear Algebra, Electromagnetism, Operations Research, Machine Learning, Power Electronics, Signals & Systems

EXPERIENCE

Embedded Systems Intern — *ADAS SAFE, Jacksonville, FL* **Jun – Aug 2025**

- Engineered real-time multi-sensor validation firmware for safety-critical ADAS systems fusing synchronized radar, IMU, and LiDAR data streams, cutting hardware alignment errors by ~15% across a production automotive sensor suite deployment
- Built a cross-sensor calibration analysis framework, extracting temporal drift patterns and inter-sensor timing dependencies from raw radar, IMU, and LiDAR logs, directly informing sensor fusion hardware architecture and OEM calibration strategy
- Automated ADAS diagnostic workflows using statistical process control over multi-modal time-series sensor streams, detecting calibration drift and misalignment early, eliminating manual hardware QA overhead across sensor pipeline

Robotics & Hardware Systems Lead — *Engineers Without Borders, Ithaca, NY* **Oct 2024 – Jan 2026**

- Adapted NASA's open-source Mars rover design for agricultural deployment, built PCB electrical subsystems, integrating a ROS-controlled camera-mounted robotic arm, and designed motor driver circuits for terrain-adaptive drive control
- Deployed a complementary autonomous drone system integrating open-source autopilot firmware, custom image-capture pipelines, and GPS-guided waypoint navigation for multi-angle aerial crop surveillance across 3 active agricultural field sites
- Trained and optimized CNN inference models deployed on NVIDIA Jetson edge hardware, achieving 40% improvement in maize disease detection accuracy with real-time GPIO-triggered farmer alerts over an embedded Linux runtime

Embedded Data Systems Intern — *Simulacrum, NYC, NY* **Jan 2026 – Present**

- Engineered a distributed low-latency embedded data platform using FastAPI and Pub/Sub, processing high-throughput real-time sensor streams with sub-100ms end-to-end latency, serving production hardware forecasting systems
- Built a production hardware observability system detecting sensor input drift, latency spikes, and pipeline failures in real time using structured logging and automated alerting, reducing embedded system debugging time by ~30%
- Designed end-to-end hardware-adjacent system architecture spanning data ingestion, model serving, and BigQuery storage with schema evolution, enabling fully reproducible ML experiments across 50+ training runs on production hardware

Systems Automation Lead & Treasurer — *Cornell Computer Reuse Association, Ithaca, NY* **Oct 2024 – Present**

- Built automated hardware triage and diagnostics tooling standardizing PCB-level fault classification, reliability scoring, and functional validation workflows, enabling scalable refurbishment and global redeployment of 350+ devices to 15 countries
- Applied systematic hardware debugging techniques, multimeter probing, drive cloning, and component-level replacement, to diagnose and restore failed laptops and desktops, sustaining a global electronics redistribution operation across communities

PROJECTS

Electrical Systems Engineer — *Cornell Engineering in Action (CU EIA)* **Sep 2024 – Aug 2025**

- Designed a solar-powered electrical system for groundwater pumping and UV disinfection serving 180 students at Matfuntini Primary School, Eswatini, sizing power distribution, wiring schematics, and load calculations in AutoCAD and MATLAB
- Developed automated pump control and disinfection logic using embedded software and sensor-driven automation, ensuring consistent water treatment output and system fault detection across fully deployed field hardware in rural Eswatini
- Conducted on-site field surveys and authored technical design reports covering electrical load analysis, component specifications, and system validation, contributing to a fully operational, deployed system serving 180-student primary school

FPGA Music Player — *Digital Logic & Computer Organization* **Nov – Dec 2025**

- Designed a fully functional FPGA-based music player in Verilog, implementing a multi-state Moore FSM architecture with clock-synchronized tone generation, achieving sub-millisecond audio output latency, and verified frequency accuracy
- Performed full timing simulation and hardware debugging in Vivado, resolving critical path violations and setup/hold time failures, ensuring stable 100 MHz clock operation with verified signal integrity across all logic paths
- Validated synthesized RTL against behavioral simulation testbenches, confirming correct FSM state transitions, output waveforms, and timing margins across all operating conditions, demonstrating full hardware design verification

SKILLS

Hardware & HDL: Verilog, VHDL, FPGA, Vivado, Quartus, digital logic design, FSM, timing analysis, synthesis, signal integrity
Embedded & Protocols: C/C++, Python, Bash, UART, I2C, SPI, PWM, JTAG, GPIO, RTOS concepts, bare-metal firmware
Robotics & Sensors: ROS/ROS2, sensor fusion, LiDAR, IMU, Raspberry Pi, Arduino, NVIDIA Jetson Orin Nano, Roboclaw
EE Tools & Design: AutoCAD, Fusion 360, oscilloscope, soldering, PCB assembly, MATLAB, Simulink, GCP, Git/GitHub
Programs: AMD AI Developer Program, BCG Launch Program, Expedition EY, Thrive NextGen Consultants